Proposed Manuscript Insertions III
End Rescaling, Adaptive Plumbing Response, and Finite-Horizon Transfer

Research draft for the marked-width MLC program

July 24, 2026

Abstract

We construct a canonical direct-limit rescaling surface for every infinite address of the first direct-decoration pullback system and prove that every finite, shift-consistent diagram on these surfaces has an exact realization by finite polynomial cylinders. We then use the source ray/equipotential quadrilaterals to obtain linear serial resistance growth and an explicit operator-norm estimate comparing a deep physical response matrix with its nodal normalization. Any preassigned additive plumbing error can be achieved by a finite depth choice. Consequently, a source-licensed strict root transition transfers through a direct-decoration virtual end with the same homogeneous contraction coefficient. These results do not identify the normalized fibers with Dudko's proposed actual planar Cantor family and do not prove MLC.

1 Direct-decoration data

Let P be a closed Jordan puzzle disk, let $g = f^q$, and let $\{Q_a : a \in \mathcal{A}\}$ be finitely many closed Jordan pullbacks such that $g : Q_a \to P$ is conformal. Write $\phi_a : P \to Q_a$ for the inverse branch. For $w = a_1 \cdots a_m$, put

$$\phi_w = \phi_{a_1} \circ \cdots \circ \phi_{a_m}, \quad Q_w = \phi_w(P), \quad h_w = g^m|_{Q_w}.$$

Let

$$F = f^r|_A : A \to B, \qquad K \Subset A \Subset B \Subset \operatorname{int} P$$

be a connected quadratic-like restriction, and let $\jmath : A \hookrightarrow B$ be inclusion. For a finite word w, define

$$A_w = h_w^{-1}(A), \quad K_w = h_w^{-1}(K),$$

$$G_w = F \circ h_w = f^{q|w|+r}|_{A_w}, \qquad I_w = \jmath \circ h_w = f^{q|w|}|_{A_w}.$$

The relative map is the root map:

$$G_w \circ I_w^{-1} = F \circ \jmath^{-1}.$$

Thus the common univalent prefix does not alter the relative critical word.

2 Canonical end rescaling

Fix $\omega = a_1 a_2 \cdots \in \mathcal{A}^{\mathbb{N}}$ and write $w_m = \omega|m$. Define the geometric-prefix map

$$t_m = h_{w_{m+1}}^{-1} \circ h_{w_m} : A_{w_m} \to A_{w_{m+1}}.$$

Then

$$h_{w_{m+1}} t_m = h_{w_m}, \qquad G_{w_{m+1}} t_m = G_{w_m}, \qquad I_{w_{m+1}} t_m = I_{w_m}.$$

Theorem 2.1 (Canonical end-rescaling surface)**.** *The direct limit*

$$\widehat{A}_\omega = \varinjlim (A_{w_m}, t_m)$$

is a Riemann surface equipped with a biholomorphism

$$h_\omega : \widehat{A}_\omega \to A.$$

The finite pairs descend to

$$\widehat{G}_\omega = F \circ h_\omega, \qquad \widehat{I}_\omega = \jmath \circ h_\omega.$$

The direct-limit carrier $\widehat{K}_\omega$ is mapped homeomorphically to K. Every finite marked datum transported by the maps h_{w_m} is exactly the corresponding root datum.

Proof. Take the quotient of $\bigsqcup_m A_{w_m}$ by the equivalence generated by $z \sim t_m(z)$. Since every transition is conformal, the quotient has the unique Riemann-surface structure making the level maps conformal. The formula $h_\omega([z, m]) = h_{w_m}(z)$ is well-defined and gives a conformal bijection onto A. Compatibility of the finite pairs and markings gives the remaining claims. □

Remark 2.2 (Scope). The theorem constructs a normalized rescaling pro-object from actual finite pullbacks. It does not identify the planar impression $E_\omega = \bigcap_m Q_{\omega|m}$ with a nondegenerate small Julia continuum.

2.1 Dynamic shift

At finite levels,

$$g : A_{a_1 \cdots a_{m+1}} \to A_{a_2 \cdots a_{m+1}}, \qquad h_{a_2 \cdots a_{m+1}} g = h_{a_1 \cdots a_{m+1}}.$$

Theorem 2.3 (End-shift naturality)**.** *The finite maps induce a biholomorphism*

$$\widehat{S}_\omega : \widehat{A}_\omega \to \widehat{A}_{\sigma\omega}$$

with

$$h_{\sigma\omega} \widehat{S}_\omega = h_\omega.$$

Hence the dynamic shift is the identity in normalized coordinates and commutes with the descended covering/immersion pair and all transported finite markings.

Warning 2.4. The dynamic shift is induced by the actual iterate f^q. The geometric-prefix map t_m is a conformal presentation change. They become equal only after normalization and must remain separately typed in the proof ledger.

3 Finite physical realization

A finite normalized diagram is called *depth graded* if its nodes carry integers $s(v)$ such that root, presentation, and enlargement edges preserve s, while a forward shift edge $u \to v$ satisfies $s(u) = s(v) + 1$.

Theorem 3.1 (Exact realization of a depth-graded diagram)**.** *Let $\mathcal{D}$ be a finite depth-graded normalized diagram whose root restrictions and marked squares are actual. For all sufficiently large M, represent node v at physical depth $m(v) = M + s(v)$. Then:*

1. *root branches are represented by the retimed pairs $(f^{qm(v)+r_e}, f^{qm(v)})$;*
2. *shift edges are represented by the actual map f^q;*
3. *every purely dynamical path is a genuine polynomial iterate on an actual nonempty domain, with the exact chronological critical word;*
4. *presentation changes remain explicit conformal coordinate maps;*
5. *all declared component, sheet, obstacle, localization, prime-end, relative-cover, response, Value, and enlargement data commute exactly.*

Proof. Pull every root object back by the cylinder chart at depth $m(v)$. The grading makes each shift source exactly one level deeper than its target. The formulas for t_m and the finite shift give all generating squares. The critical-spine telescope and marked finite-horizon assembly apply to the dynamical compositions. Finiteness permits one common M. □

Remark 3.2. If two paths into one node have different net shift counts, the diagram must be unfolded by duplicating that node. Acyclicity alone does not imply physical depth consistency.

4 Intrinsic finite pseudo-puzzle atlas

Let $H : U \to V$ be an externally marked connected quadratic-like map. Fix a finite source-authorized list of rational external accesses, equipotential cross-sections, and pullback depths. Construct the finite access/equipotential graph in the prime-end external class and take its actual pullbacks under H.

Theorem 4.1 (Intrinsic finite atlas)**.** *The resulting finite pseudo-puzzle consists of actual component domains and actual restrictions of iterates H^n. It retains exact return times, total and local degrees, chronological critical words, component and sheet routes, obstacles, prime-end phase, and cyclic order. It is functorial under an externally marked conjugacy of the declared finite atlas.*

Proof. The external marking fixes the phase of the degree-two external covering. The finite graph is therefore defined in the external prime-end class. Pullbacks, components, and return exclusions are read from actual iterates of H. No uniqueness of an arbitrary hybrid conjugacy is used. □

Corollary 4.2 (Finite recursive compiler)**.** *Every finite source-supplied decorated chain of externally marked connected quadratic-like maps admits a prefix-local recursive realization on the normalized end surfaces. Every finite multi-scale diagram has an exact finite physical realization. The construction is covariant under marked changes of presentation and does not assume rigidity.*

5 Exact fiberwise geometry

Let $X \subset A \setminus K$ be a marked complement localization, let Γ be a declared curve family, and transport all curves and electrodes by h_w and h_ω.

Theorem 5.1 (Exact width and response)**.** *For every finite word and infinite address,*

$$\mathrm{Ext}_{X_w}(h_w^{-1}\Gamma) = \mathrm{Ext}_X(\Gamma) = \mathrm{Ext}_{\widehat{X}_\omega}(h_\omega^{-1}\Gamma).$$

If Λ_X is the finite-electrode Dirichlet-to-Neumann matrix, then

$$\Lambda_{X_w} = \Lambda_X = \Lambda_{\widehat{X}_\omega}.$$

Parallel gluing and Schur-complement elimination commute with these equalities.

Proof. Extremal length and two-dimensional Dirichlet energy are conformal invariants. The marked maps preserve electrode order and orientation. The response gluing operations are algebraic. □

Corollary 5.2. *Any finite WAD, response, or Value state whose entire defining incidence, frames, coefficient rows, and future observations are transported by the marked conformal map is an identity state along the normalized address tail.*

6 Serial ray/equipotential necks

For each letter a, choose a source-authorized quadrilateral $C_a \subset P \setminus \operatorname{int} Q_a$ whose horizontal sides are parent and child equipotential cross-sections and whose vertical sides are the two retained ray/access walls. Let $\mathcal{V}_a$ join the horizontal sides and put

$$R_a = \mathrm{Ext}_{C_a}(\mathcal{V}_a) > 0, \qquad R_* = \min_{a\in\mathcal{A}} R_a > 0.$$

For $w = a_1 \cdots a_m$, set

$$C_{w,j} = \phi_{a_1\cdots a_{j-1}}(C_{a_j}).$$

Call the localization serially exhaustive if every authorized deep-to-parent curve contains a crossing of every $C_{w,j}$ in order.

Theorem 6.1 (Serial resistance)**.** *Under serial exhaustion, the effective resistance and conductance satisfy*

$$R(w) \geq \sum_{j=1}^{m} R_{a_j} \geq mR_*,$$

$$c(w) \leq \left(\sum_{j=1}^{m} R_{a_j}\right)^{-1} \leq \frac{1}{mR_*}.$$

Proof. The pulled cells have disjoint interiors, are conformally identical to their root cells, and are crossed in series. Apply the extremal-length Series Law. □

Remark 6.2. No block-independent lower bound for R_* is needed below. The physical materialization depth adapts to the actual positive value.

7 Adaptive nodal response

Let $S = \{\omega_1, \dots, \omega_s\}$ be the end addresses read by a finite proof horizon. Choose a depth beyond their common prefixes and, for address i, an additional serial tail of length ℓ_i. Cut across the parent-side end of each disjoint tail. Let Λ_{phys} be the connected response and Λ_{node} the block-diagonal response of the cut normalization. Set

$$\mathcal{R}_i = \sum_{j=1}^{\ell_i} R_{i,j}, \qquad c_i = \mathcal{R}_i^{-1}.$$

Theorem 7.1 (Quantitative nodal-response approximation)**.** *On the quotient by componentwise constants,*

$$0 \preceq \Lambda_{\mathrm{phys}} - \Lambda_{\mathrm{node}}.$$

For every terminal vector x,

$$0 \le x^T(\Lambda_{\mathrm{phys}} - \Lambda_{\mathrm{node}})x \le \left(\sum_{i=1}^{s} c_i\right) \operatorname{osc}(x)^2.$$

Consequently,

$$\|\Lambda_{\mathrm{phys}} - \Lambda_{\mathrm{node}}\|_{\mathrm{op}} \le 2\sum_{i=1}^{s} c_i \le 2\sum_{i=1}^{s} \frac{1}{\ell_i R_{*,i}}.$$

Proof. Cutting removes nonnegative coupling energy and permits independent constants on the normalization components, proving the lower inequality. Take the energy minimizers on the cut components. By the maximum principle, their free attachment values lie between the global minimum and maximum terminal values. Interpolate through neck i by its two-terminal energy minimizer. This costs at most $c_i \operatorname{osc}(x)^2$. Summing gives the quadratic-form bound. Finally, $\operatorname{osc}(x)^2 \le 2\|x\|_2^2$. □

Corollary 7.2 (Explicit depth choice)**.** *For a prescribed tolerance $\varepsilon > 0$, it suffices to choose*

$$\sum_i \frac{2}{\ell_i R_{*,i}} \le \varepsilon.$$

If one lower bound R_* applies to all selected tails, the uniform choice

$$\ell_i \ge \left\lceil \frac{2s}{\varepsilon R_*} \right\rceil$$

gives $\|\Lambda_{\mathrm{phys}} - \Lambda_{\mathrm{node}}\|_{\mathrm{op}} \le \varepsilon$.

Remark 7.3. The depth is finite side data selected after the finite proof horizon is frozen and from a predeclared error budget. The normalized end object itself does not depend on this choice.

8 Finite-horizon and transition consequences

Assume a finite normalized diagram has actual intrinsic branches, one quadratic relative stage, critical-spine retention, a protected root packet and port certificate, serially exhaustive necks, and a prescribed response error.

Theorem 8.1 (Direct-decoration finite-horizon inflation)**.** *The diagram has an actual finite physical realization with locally saturated pseudo-quadratic-like stages, selected relative degree two, exact chronological critical word and complete markings, a protected finite carrier, exact fiberwise response modules, and total end-plumbing response error below the prescribed tolerance.*

Proof. Use the depth-graded realization theorem, the critical-spine telescope and marked assembly, the finite protected-carrier theorem, exact fiberwise response, and the adaptive depth corollary. □

Let L be an error-stable homogeneous gauge satisfying

$$L(Z+\Delta) \le L(Z) + C_L\|\Delta\|.$$

Assume the root block satisfies

$$L(Z_{\mathrm{out}}) \le \mu L(Z_{\mathrm{in}}) + E_{\mathrm{root}}, \qquad 0 < \mu < 1.$$

Theorem 8.2 (Strict transition transfer)**.** *For any preassigned $E_{\mathrm{pl}} > 0$, choose the materialization depth so that $\|\Delta_{\mathrm{pl}}\| \le E_{\mathrm{pl}}/C_L$. Then*

$$L(Z^{\mathrm{vir}}_{\mathrm{out}}) \le \mu L(Z^{\mathrm{vir}}_{\mathrm{in}}) + E_{\mathrm{root}} + E_{\mathrm{pl}}.$$

Equivalently,

$$L_{\mathrm{in}} \ge \mu^{-1} L_{\mathrm{out}} - \mu^{-1}(E_{\mathrm{root}} + E_{\mathrm{pl}}).$$

The homogeneous backward factor $\lambda = \mu^{-1} > 1$ is unchanged by the virtual address tail.

Proof. The normalized univalent tail is an identity chart. The only added response term is the plumbing perturbation, controlled by the preceding theorem. Apply error stability and rearrange. □

8.1 Planar-free finite-interface execution

Call a block finite-interface when every proof call reads a finite depth-graded diagram; end operations are conformally natural; cross-fiber interaction is read only through a finite protected carrier, finite-electrode response, and retained finite Value features; inequalities are stable under a prescribed response error; and the horizon and error budget are frozen before depth is chosen.

Theorem 8.3 (Pro-exact execution sufficiency)**.** *A finite-interface direct-decoration block supplied with its actual root branches, intrinsic finite atlases, critical spine, packet/port certificate, and serial necks has, for every $E_{\mathrm{pl}} > 0$, an actual finite-cylinder execution with exact dynamical and marked data, one protected finite carrier, and total cross-fiber response error at most E_{pl}. Every error-stable strict block estimate survives with that additive charge. No point-valued landing map or nondegenerate planar infinite-address component is used.*

Proof. Unfold shift-grading conflicts and apply finite physical realization. Use the protected-carrier theorem, distribute the finite error budget among the selected tails, and choose their depths by the nodal-response estimate. Exact fiber data and strict-transition transfer finish the execution. □

Thus planar virtual-component identification is not a prerequisite for any module satisfying this finite-interface contract. Universal source authorization and a uniformly adequate observable interface remain open.

9 Promotion boundary

The following split is required to avoid theorem laundering:

Original node	Proved part	Still open
Disconnected marking transfer	Rescaled end transfer and finite realization	Identification with the intended planar virtual component
Cantor package realization	Fibered pro-exact package	Actual planar Cantor family and global protected coupling
Prefix-local scale selector	Intrinsic finite atlas compiler	Universal source authorization at every mixed scale

The present results also do not prove a universally bounded future-observable response/WAD/Value dimension, a strict source alternative for every mixed block, MBTT, coercive compactness, or mixed-tower rigidity.

10 Next theorem

The next source-facing theorem should universally authorize the finite normalized interface for every mixed block: actual root portraits, intrinsic finite atlases, critical spines, protected packet/port and serial-neck certificates, and a proof-complete demand schema. It should then construct a universally controlled future-observable response/WAD/Value quotient of the complete coupled interaction, with a strict root alternative and discounted error budget. The planar identification of infinite-address components remains an open geometric refinement, but Theorem 8.3 shows that it is not a logical prerequisite for this finite-interface route. The remaining global steps are MBTT, geometric coercivity, and mixed-tower rigidity.